

SearchJev: A Fast and Calibrated System-1 Model for Search Agents

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 Code  Models

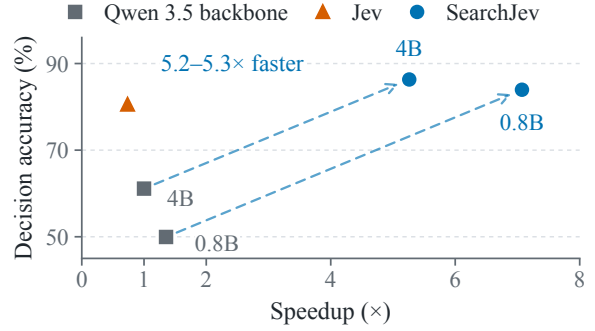
Abstract

Search agents repeatedly make short decisions about relevance, evidence sufficiency, and search actions. Using generative language models for these decisions introduces latency and unreliable confidence. We present SEARCHJEV, a fast and calibrated System-1 model that separates search decisions from System-2 reasoning and generation. Given a search state and a decision schema, SEARCHJEV directly scores legal options without autoregressive output generation. We propose Soft-Label Learning for Calibrated Decisions (SLCD) to learn decision probabilities from uncertain supervision and calibrate their confidence. In a dual-system search agent, SEARCHJEV handles short decisions and delegates uncertain judgments to System 2, which retains planning, query generation, and answer composition. We also introduce SearchDecision-Bench, a benchmark unifying six types of search decisions for training and evaluation. On SearchDecision-Bench, SEARCHJEV improves decision quality over same-size Qwen3.5 autoregressive models, achieves $5.2\text{--}5.3\times$ faster decisions, and reduces average expected calibration error by 41-74%. On BrowseComp-Plus, the dual-system agents achieve a $3.7\text{--}4.7\times$ speedup in active search time while improving answer accuracy from 45% to up to 54%.

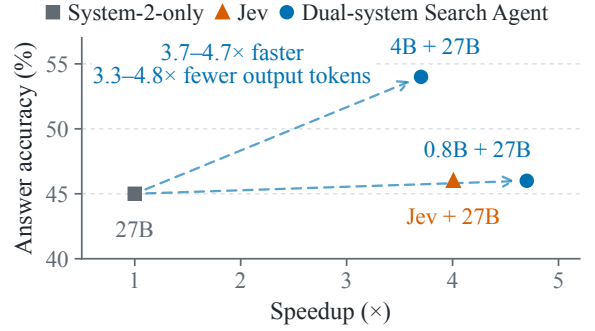
1 Introduction

LLM-based search agents interleave reasoning with retrieval to answer complex questions (Yao et al., 2023; Jin et al., 2025; Li et al., 2025). Within these trajectories, agents repeatedly make short decisions: *Is this passage relevant? Is the evidence sufficient to answer? Which link should be explored next?* These judgments determine which evidence is used and how the search proceeds. Drawing on the distinction between System 1 and System 2 (Kahneman, 2011), we view them as *System-1* decisions

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(a) Search decision performance.



(b) End-to-end agentic search.

Figure 1: Search decision performance and end-to-end agentic search. (a) On SearchDecision-Bench, SEARCHJEV achieves $5.2\text{--}5.3\times$ faster decisions than same-size Qwen3.5. The x-axis uses Qwen3.5-4B as the $1\times$ reference. (b) On BrowseComp-Plus, dual-system agents with SEARCHJEV-0.8B/4B reach 46.0%/54.0% answer accuracy, compared with 45.0% for System-2-only and 46.0% for Jev 1.13. Relative to System-2-only, they achieve $3.7\text{--}4.7\times$ faster search with $3.3\text{--}4.8\times$ fewer output tokens.

alongside the *System-2* reasoning needed for planning and answer generation. Handling both through the same generative process adds sequential decoding overhead to frequent search decisions.

Two common approaches illustrate the difficulty of unifying search decisions. Task-specific rerankers (Nogueira et al., 2020) and query-complexity classifiers (Jeong et al., 2024) support individual functions, but require separate models

and supervision for different decisions. Prompting a general LLM accommodates diverse tasks, but uses autoregressive generation to produce short judgments. Its confidence, whether verbalized or derived from token probabilities, is often miscalibrated (Kadavath et al., 2022; Tian et al., 2023; Xiong et al., 2024). Reliable confidence is needed to determine whether a decision can be accepted or requires further reasoning. The challenge is therefore to handle heterogeneous search decisions with a unified model that is both efficient and well calibrated.

To address this challenge, we propose SEARCHJEV, a fast and calibrated System-1 model for search agents. Inspired by Jev’s use of typed probabilistic outputs for automation (Almeida, 2026), our central idea is to learn search judgments as schema-conditioned probabilistic predictions and separate them from System-2 reasoning and generation. This separation allows one model to handle multiple decision types and provides confidence estimates for deciding when to invoke System 2 for a judgment.

SEARCHJEV has three components. (i) The *System-1 Search Decision Model* uses a textual schema to specify each decision and its legal options. Given the search state and schema, it normalizes the LM’s first-output-token logits over legal option labels to directly produce a decision distribution, avoiding autoregressive output generation. (ii) *Soft-Label Learning for Calibrated Decisions* (SLCD) learns these distributions from soft targets that represent uncertainty in the supervision. It combines proper scoring losses with subsequent temperature calibration to improve the reliability of decision confidence (Gneiting and Raftery, 2007; Guo et al., 2017). (iii) The *Dual-System Search Agent* uses SEARCHJEV for short search decisions and delegates uncertain judgments to System 2, which retains planning, query generation, and answer composition. This division of labor replaces repeated generative judgments with direct predictions while preserving access to System-2 reasoning.

To support unified training and evaluation, we introduce SearchDecision-Bench. Existing supervision for search decisions comes from heterogeneous resources with different input and label formats. We convert these resources into a common state-schema format covering six decision types: routing, rewriting, relevance, sufficiency, navigation, and verification. The benchmark evaluates

decision quality, calibration, and latency, and includes held-out datasets and splits for assessing generalization.

We evaluate SEARCHJEV at both the decision and agent levels. On SearchDecision-Bench, SEARCHJEV improves decision quality and reduces average calibration error relative to same-size Qwen3.5 AR (JSON) models, while achieving $5.2\text{--}5.3\times$ faster decisions. On BrowseCompPlus (Chen et al., 2026), the dual-system agents achieve a $3.7\text{--}4.7\times$ speedup in median active time while improving answer accuracy from 45% to up to 54%, compared with the System-2-only agent. Figure 1 summarizes these gains in search decision performance and end-to-end agentic search.

Our contributions are:

- We propose SEARCHJEV, a schema-conditioned search decision model that directly predicts legal options and learns calibrated probabilities through SLCD.
- We build a dual-system search agent that separates short decisions from reasoning and generation, and delegates uncertain judgments from System 1 to System 2.
- We formalize six types of search decisions and introduce SearchDecision-Bench for unified training and evaluation.
- We demonstrate improved decision quality, average calibration, and efficiency over same-size autoregressive models, and better end-to-end answer accuracy with lower search time than a System-2-only agent.

2 SearchDecision-Bench

We introduce SearchDecision-Bench to train and evaluate short decisions made during agentic search. It unifies heterogeneous search resources as state-schema pairs with field-level targets, covering six decision types and three output types.

2.1 Task Definition and Taxonomy

Task definition. Given a search state and a decision schema, the task is to predict a legal option for each requested decision and estimate its confidence. At step t , the state $s_t = (q, H_t, C_t)$ contains the user request q , the trajectory H_t of queries, opened documents, and intermediate notes, and candidates $C_t = \{c_1, \dots, c_n\}$ such as documents, links, or sources. A schema $\sigma = \{f_1, \dots, f_k\}$ specifies the decisions as named fields, each with a natural-language question and a set of legal options $\mathcal{V}(f)$.

Decision Type	Decision Task	Output Type	Task Example
Routing	Intent classification	Choice	Query: <i>{query}</i> . Which intent does it express? Options: <i>{intent labels}</i> .
	Retrieval need	Score	Query: <i>{query}</i> . How much information must be retrieved? Levels: none / one fact / several facts / multi-source research.
	Query clarity	Choice	Query: <i>{query}</i> . What information most needs to be added? Options: none / time / location / object / aspect / <i>{other}</i> .
Rewriting	Query equivalence	Noul	Queries: <i>{query 1}</i> and <i>{query 2}</i> . Do they express the same intent? Options: True / False.
	Rewrite equivalence	Noul	Dialogue: <i>{history}</i> . Utterance: <i>{utterance}</i> . Rewrite: <i>{rewrite}</i> . Does the rewrite keep the utterance’s intent? Options: True / False.
Relevance	Relevance judgment	Noul / Score	Query: <i>{query}</i> . Document: <i>{document}</i> . Is it relevant? Options: True / False, or <i>{relevance grades}</i> .
	Document usefulness	Noul	Sub-query: <i>{current sub-query}</i> . Document: <i>{document}</i> . Does the document answer the sub-query? Options: True / False.
	Document quality	Score	Query: <i>{query}</i> . Document: <i>{site, date, content}</i> . How authoritative is the source? Levels: untrusted / user-generated / professional / authoritative.
Sufficiency	Evidence sufficiency	Noul	Question: <i>{original question}</i> . Evidence: <i>{retrieved evidence}</i> . Does the evidence cover all facts needed to answer? Options: True / False.
Navigation	Next-action selection	Choice	Search state: <i>{state}</i> . What should the agent do next? Options: search / read page / answer.
	Next sub-query selection	Choice	Question: <i>{original question}</i> . Solved steps: <i>{history}</i> . Which sub-query comes next? Options: <i>{four candidate queries}</i> .
	Query-fit assessment	Noul	Next unsolved step: <i>{step}</i> . Candidate query: <i>{query}</i> . Does the query target this step? Options: True / False.
Verification	Claim verification	Choice	Claim: <i>{claim}</i> . Evidence: <i>{evidence}</i> . Is the claim supported? Options: supports / refutes / insufficient.
	Evidence detection	Noul	Claim: <i>{claim}</i> . Sentence: <i>{sentence}</i> . Is the sentence evidence for the claim? Options: True / False.

Table 1: Representative decision tasks for agentic search in SearchDecision-Bench. The illustrative examples span six decision types and three output types: Choice (categorical selection), Score (ordinal scoring), and Noul (binary judgment). Braces denote placeholders.

Decision Type	Training	ID Test	OOD Test
Routing	77,627	26,427	10,000
Rewriting	74,318	32,746	10,000
Relevance	153,990	17,581	5,976
Sufficiency	14,652	1,036	0
Navigation	56,981	10,213	0
Verification	47,830	10,334	20,631

Table 2: Statistics of SearchDecision-Bench. Counts denote state-schema pairs.

A sample consists of a serialized state, a schema, and a target for each field; several fields can share the same state. We write $s = s_t$ when the step index is not needed.

Decision types. We organize search decisions into six types: *routing* identifies a query’s intent or retrieval needs; *rewriting* checks a candidate query rewrite; *relevance* judges retrieved documents; *sufficiency* determines whether the available evidence is enough; *navigation* selects or assesses the next

search step; and *verification* checks whether evidence supports a claim. Table 1 presents representative tasks and examples for each type.

Output types. SearchDecision-Bench uses *Choice* for categorical selection, *Score* for ordinal scoring, and *Noul* for binary judgments. Legal options are specified in text; a decision type can use multiple output types.

2.2 Data Construction and Splits

Data sources. We draw on Chinese and English resources that provide supervision for complementary search decisions. Query-understanding and rewriting datasets, such as BANKING77 (Casanueva et al., 2020), LCQMC (Liu et al., 2018), and QReCC (Anantha et al., 2021), support intent classification and query-equivalence decisions. Retrieval datasets, including T2Ranking (Xie et al., 2023) and

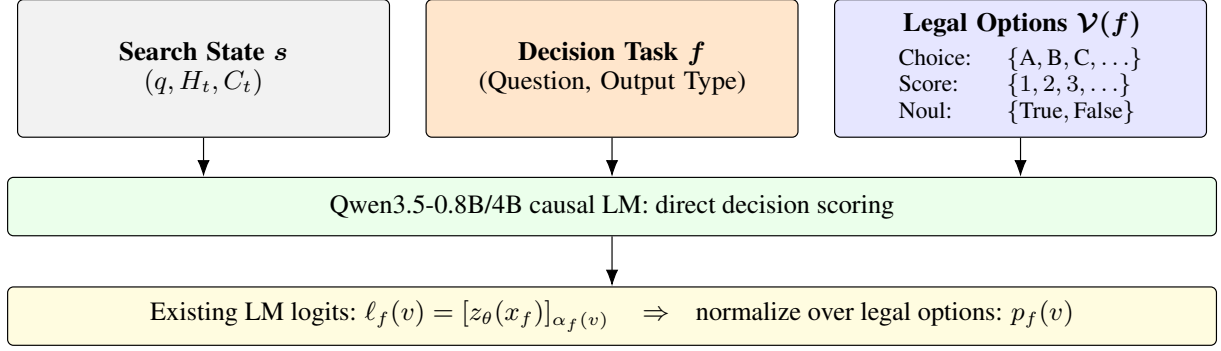


Figure 2: SEARCHJEV decision model. A search state, the question and output type of task f , and its legal options form the decision input x_f . Choice uses letter labels; Score and Noul show example values. The model normalizes option-label logits into a decision distribution without autoregressive output generation.

DuReader-Retrieval (Qiu et al., 2022), provide passage-level judgments. Multi-hop QA and agent demonstrations support sufficiency and navigation decisions, while claim-evidence resources such as VitaminC (Schuster et al., 2021) provide verification labels. Training also includes natural language inference, sentiment analysis, topic classification, and reading comprehension data (Xu et al., 2020; Tan and Zhang, 2008).

Sample construction and annotation. We convert each source instance into a search state and instantiate decision fields with textual questions and legal options. States range from query-passage, query-query, and claim-evidence pairs to multi-hop states built from MuSiQue (Trivedi et al., 2022) decompositions. Supervision comes from original annotations, derived labels, and LLM teachers or votes. For example, MuSiQue decompositions and hard negatives provide evidence-sufficiency and next-step judgments, filtered by an LLM; agent demonstrations provide next-action labels. We encode labels as soft targets, assigning confidence according to the supervision source, and paraphrase field descriptions in Chinese or English to express the same decision through different instructions.

Quality control. We deduplicate identical inputs, remove samples with conflicting labels, and mask fields that cannot be judged from the available information. We isolate splits using identity keys and query text, and discard synthetic variants whose source appears in an evaluation split. Training and validation inputs whose states also occur in a test set are removed.

Splits and statistics. In-distribution (ID) tests use held-out samples from the same datasets used

for training. Out-of-distribution (OOD) evaluation uses held-out datasets and splits: FRAMES (Krishna et al., 2025), RAGTruth (Niu et al., 2024), CANDY (Guo et al., 2025), Search Arena (Miroyan et al., 2026), AgentRewardBench (Lù et al., 2025), and the QReCC test split. Both ID and OOD test samples are excluded from training. Table 2 counts state-schema pairs for the six search-decision types: all six have ID tests, and four have OOD tests. The general-language training tasks are outside these counts. BrowseComp-Plus (Chen et al., 2026) is used only for separate end-to-end agentic search evaluation.

3 SEARCHJEV

SEARCHJEV separates short search decisions from the reasoning and generation performed by System 2. A token-logit readout directly predicts a distribution over legal options without autoregressive output generation. Soft-label learning trains these probabilities, followed by confidence calibration. The dual-system agent then uses SEARCHJEV for search decisions, delegating uncertain cases to System 2 while retaining System 2 for planning, query generation, and answers.

3.1 System-1 Search Decision Model

Short search decisions often require choosing among a small set of legal options. SEARCHJEV reads their probabilities directly from a pre-trained causal LM’s output logits, avoiding multi-token structured generation. Figure 2 illustrates the model.

Schema-conditioned input. For each decision task $f \in \sigma$, the input x_f combines the search state s , the task’s question and output type, and its le-

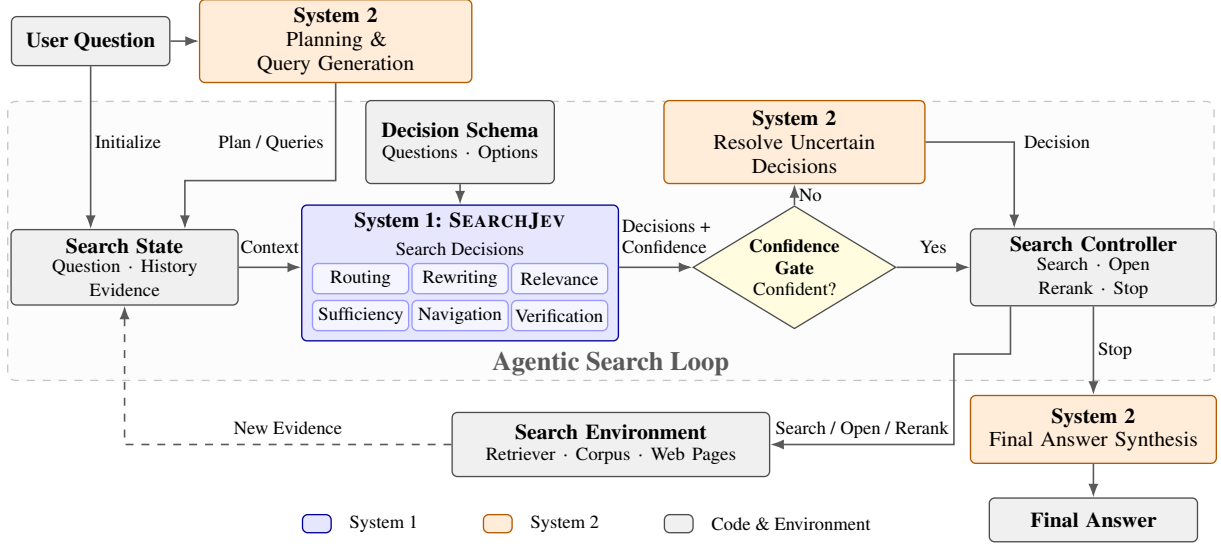


Figure 3: Architecture of the Dual-System Search Agent. System 2 handles planning, query generation, and final answer synthesis, while SEARCHJEV (System 1) makes search decisions specified by the schema at each step. Confident decisions are accepted; uncertain ones are delegated to System 2. The search controller uses the resolved decisions to select actions and update the search state with retrieved evidence.

gal options $\mathcal{V}(f)$. The schema specifies a finite option set for each task. Each option is paired with a distinct single-token label from the existing vocabulary, such as A or B. The input ends with an instruction to select one label and the assistant generation prefix. Option descriptions can span multiple tokens; only their labels must be single tokens.

Token-logit decision readout. We use the LM’s existing vocabulary projection to score the legal labels. Let $z_\theta(x_f)$ be the vocabulary-logit vector at the first output position, where θ denotes the model parameters. The token index $\alpha_f(v)$ identifies the label assigned to option v . The option score and decision probability are

$$\begin{aligned} \ell_f(v) &= [z_\theta(x_f)]_{\alpha_f(v)}, \\ p_f(v) &= \frac{\exp \ell_f(v)}{\sum_{u \in \mathcal{V}(f)} \exp \ell_f(u)}, \end{aligned} \quad (1)$$

where $v \in \mathcal{V}(f)$ and $p_f(v)$ abbreviates the model’s uncalibrated probability $p_\theta(v \mid x_f)$. Normalizing only over legal labels gives a distribution for Choice, Score, or Noul decisions and removes the need for output parsing. The readout uses the existing LM head without adding decision-specific parameters.

Direct decision inference. Reading legal-label logits produces decision distributions without generating structured responses token by token. De-

cision inputs can be processed in batches, and the label-to-option mapping supplies structured outputs without JSON parsing. This avoids the sequential decoding overhead of autoregressive decision models while retaining the Qwen3.5 backbone (Qwen Team, 2026).

3.2 Soft-Label Learning for Calibrated Decisions

Search decisions require informative confidence estimates, but their supervision comes from sources with different reliability. We propose *Soft-Label Learning for Calibrated Decisions* (SLCD) to learn decision probabilities from soft targets and calibrate them by output type.

Soft-label supervision. SLCD uses a target distribution t_f over $\mathcal{V}(f)$ to represent uncertainty in its supervision. It uses the soft targets supplied by the data construction process, with one-hot targets for labels without a probability distribution. We paraphrase field descriptions in Chinese or English to encourage learning from schema text, and permute the options of unordered fields to reduce position bias (Zheng et al., 2024; Pezeshkpour and Hruschka, 2024).

Decision probability learning. We fit each target distribution with a weighted combination of cross-

entropy and Brier losses:

$$\mathcal{L}_f = w_f \left[- \sum_{v \in \mathcal{V}(f)} t_f(v) \log p_f(v) + \lambda_B \sum_{v \in \mathcal{V}(f)} (p_f(v) - t_f(v))^2 \right], \quad (2)$$

where $\lambda_B \geq 0$ balances the losses and $w_f > 0$ downweights LLM-assigned labels, demonstrations, or constructed rewrites. Training averages $\mathcal{L}_f$ over the individual field examples in each mini-batch. Both losses are strictly proper scoring rules: for each field, their combination is uniquely minimized at its soft target distribution (Brier, 1950; Gneiting and Raftery, 2007).

Output-type temperature calibration. After training, we fit positive temperatures on held-out validation decisions by minimizing the negative log-likelihood of their target labels (Guo et al., 2017). For field f , the calibrated distribution is

$$\tilde{p}_f(v) = \frac{\exp(\ell_f(v)/T_{\kappa(f)})}{\sum_{u \in \mathcal{V}(f)} \exp(\ell_f(u)/T_{\kappa(f)})}, \quad (3)$$

where $\kappa(f)$ is the output type (Choice, Score, or Noul) and $T_{\kappa(f)} > 0$ is its fitted temperature. Types with insufficient validation examples use a temperature fitted across all types. Scaling preserves the probability ranking of legal options and provides the probabilities used by the search agent.

3.3 Dual-System Search Agent

We integrate SEARCHJEV into the agentic search loop shown in Figure 3. System 2 plans, writes search queries, and composes the final answer, while SEARCHJEV (System 1) handles schema-defined search decisions. A confidence gate delegates uncertain judgments to System 2, and a search controller converts the resolved decisions into actions.

Schema-conditioned decisions. The decision schema specifies the judgments needed at each search step. Given search state s_t and schema σ , SEARCHJEV processes the schema-conditioned inputs $\{x_f\}_{f \in \sigma}$ and returns a calibrated distribution $\tilde{p}_f$ over each field’s legal options. Changing the schema and the controller’s action rules adapts the agent to different search tasks while retaining the same division of labor.

Confidence-gated delegation. For each field f , the confidence gate accepts SEARCHJEV’s decision when its confidence $\max_{v \in \mathcal{V}(f)} \tilde{p}_f(v)$ is at least $\delta \in [0, 1]$; otherwise, System 2 resolves the same decision question using its legal options. Both paths pass their decisions to the same search controller. The threshold δ controls how many judgments are delegated to System 2 (Geifman and El-Yaniv, 2017).

Search execution and state update. The search controller applies task-specific rules to map the resolved judgments to actions such as search, open, rerank, or stop. Search and page-opening actions access the search environment, and the retrieved evidence updates the search state for subsequent decisions. When the controller stops searching, System 2 synthesizes the final answer from the accumulated evidence. Accepting confident SEARCHJEV outputs avoids extra System-2 judgment calls while retaining System 2 for planning and generation.

4 Experiments

4.1 Research Questions

We aim to answer the following research questions in our experiments: **RQ1:** How does SEARCHJEV perform in terms of decision effectiveness and efficiency? **RQ2:** How does SEARCHJEV affect the answer quality and efficiency of end-to-end agentic search? **RQ3:** How well does SEARCHJEV generalize to held-out datasets across different decision families?

4.2 Baselines

To evaluate the effectiveness of SEARCHJEV at the decision and agent levels, we compare it with the following baselines:

- **Qwen3.5 AR (JSON)** uses the same Qwen3.5-0.8B/4B backbones before task-specific training, with constrained JSON decoding on the same prompts. Confidence scores are derived from option log-probabilities.
- **Jev 1.13 (Almeida, 2026)** is a closed System-1 model, which we query through its API on the same test items. We also use Jev in place of SEARCHJEV in the dual-system agent for end-to-end comparison.
- **System-2 only** uses Qwen3.5-27B for both short decisions and search reasoning. We compare it with dual-system agents that delegate the short

Model	Relevance		Sufficiency		Routing		Navigation		Rewriting		Verification		Latency
	NDCG $\uparrow$	ECE $\downarrow$	Acc $\uparrow$	ECE $\downarrow$	Acc $\uparrow$	ECE $\downarrow$	Acc $\uparrow$	ECE $\downarrow$	Acc $\uparrow$	ECE $\downarrow$	Acc $\uparrow$	ECE $\downarrow$	
Jev 1.13 (closed)	98.5	7.6	79.8	<u>5.6</u>	73.5	<u>5.5</u>	72.1	1.5	79.5	2.8	79.8	1.6	265.4
Qwen3.5-0.8B AR (JSON)	85.7	15.7	49.9	24.2	24.7	24.0	42.0	19.7	40.6	32.1	56.8	9.5	144.2
Qwen3.5-4B AR (JSON)	96.7	4.2	57.7	15.0	53.3	6.7	37.8	26.3	66.1	5.5	55.1	<u>4.7</u>	195.2
SEARCHJEV-0.8B	<u>98.6</u>	<u>5.8</u>	<u>89.4</u>	5.1	<u>83.2</u>	5.0	56.8	<u>4.6</u>	<u>84.0</u>	<u>2.3</u>	<u>91.7</u>	9.3	27.6
SEARCHJEV-4B	99.0	6.5	96.0	<u>5.6</u>	86.6	5.7	<u>57.2</u>	8.4	85.9	2.0	93.2	8.8	<u>37.1</u>

Table 3: Main results on the in-distribution test sets of SearchDecision-Bench, covering decision quality, calibration, and latency. NDCG@10, accuracy, and ECE are scaled by 100; latency is the median per-decision time in milliseconds at batch size 1. **Bold** highlights the best performance, underlined indicates the second-best.

decisions to SEARCHJEV-0.8B or SEARCHJEV-4B.

4.3 Evaluation Metrics

To evaluate decision effectiveness on SearchDecision-Bench, we report accuracy for the five classification-based families and NDCG@10 (Järvelin and Kekäläinen, 2002) for relevance. We measure calibration using expected calibration error (ECE; Naeini et al., 2015) and efficiency using median latency per decision.

For end-to-end agentic search, we follow the answer-evaluation protocol of BrowseComp-Plus (Chen et al., 2026): an LLM judge compares the final answer with the reference answer, and accuracy is the percentage of questions judged correct. We also report System-2 calls, output and reasoning tokens, and median active time per question, including System-2 calls, decision-model inference, and retrieval.

4.4 Implementation Details

We initialize SEARCHJEV-0.8B and SEARCHJEV-4B from Qwen3.5-0.8B/4B (Qwen Team, 2026) and train them with LoRA (Hu et al., 2022) using SLCD. We set $\lambda_B = 0.5$ and assign $w_f = 0.5$ to LLM-assigned labels, demonstrations, and constructed rewrites, and $w_f = 1$ otherwise. Output-type temperatures are fitted on 5,000 validation decisions. Decision latency is measured at batch size 1 using the same inference code for the open models.

For end-to-end evaluation, we randomly sample 100 BrowseComp-Plus questions and run each agent once per question, using Qwen3.5-27B as System 2 and BM25 (Robertson and Zaragoza, 2009) as the retriever. We consider three decision types in BrowseComp-Plus: (i) *relevance* judges retrieved documents against the current query to select useful evidence; (ii) *sufficiency* determines

whether the evidence is enough to answer or further search is needed; and (iii) *verification* assesses whether the retained evidence supports the final answer, using the calibrated support probability as an answer-support score.

5 Experimental Results and Analysis

To answer our research questions, we evaluate decision effectiveness and efficiency on SearchDecision-Bench, assess end-to-end answer quality and efficiency on BrowseComp-Plus, and examine generalization to held-out datasets.

5.1 Decision Effectiveness and Efficiency (RQ1)

Table 3 reports decision quality, calibration, and latency across the six in-distribution decision families of SearchDecision-Bench. We compare SEARCHJEV-0.8B and SEARCHJEV-4B with same-size Qwen3.5 AR (JSON) models and Jev 1.13. Based on these results, we make three observations:

- **SEARCHJEV improves decision quality at both model sizes.** Both variants outperform same-size Qwen3.5 AR (JSON) across all six decision families, with the 4B model leading five of them. Compared with Jev 1.13, both improve sufficiency, routing, rewriting, and verification and achieve comparable relevance, while Jev remains stronger on navigation. These gains support our System-1 search decision design: the schema-conditioned model reads logits for legal option labels from each decision prompt, while decision-specific supervision teaches the backbone to distinguish the choices required by each schema. This aligns prediction and training with the judgments made during search.
- **SEARCHJEV improves average calibration over Qwen3.5 AR (JSON).** Across the six families, mean ECE falls by 74.4% and 40.7% relative

Agent	Acc $\uparrow$	System-2 Calls $\downarrow$	Output Tokens (k) $\downarrow$	Time (s) $\downarrow$
System-2 only (Qwen3.5-27B)	45.0	90.3	73.7	4080
Dual-system Search Agent (Jev 1.13)	<u>46.0</u>	<u>76.0</u>	<u>18.3</u>	<u>1018</u>
Dual-system Search Agent (SEARCHJEV-0.8B)	<u>46.0</u>	62.1	15.5	868
Dual-system Search Agent (SEARCHJEV-4B)	54.0	84.6	22.1	1102

Table 4: End-to-end agentic search results on BrowseComp-Plus, covering answer quality and efficiency with System 2 fixed at 27B. Dual-system agent sizes refer to SEARCHJEV; the Jev 1.13 agent uses Jev in its place. Acc is answer accuracy (%). Calls and output tokens are for System 2 and are reported per question; k denotes thousands. Time is the median active time per question, including System-2 calls, decision-model inference, and retrieval. **Bold** highlights the best performance, underlined indicates the second-best.

Model	Routing		Rewriting		Relevance		Verification		Latency
	Acc $\uparrow$	ECE $\downarrow$	Acc $\uparrow$	ECE $\downarrow$	NDCG $\uparrow$	ECE $\downarrow$	Acc $\uparrow$	ECE $\downarrow$	ms $\downarrow$
Jev 1.13 (closed)	62.3	20.0	<u>97.7</u>	5.6	<u>98.7</u>	3.4	84.5	3.7	273.3
Qwen3.5-0.8B AR (JSON)	29.2	13.4	47.3	38.4	82.2	32.0	53.9	22.6	153.2
Qwen3.5-4B AR (JSON)	43.7	30.3	95.8	<u>24.1</u>	93.8	<u>10.6</u>	<u>78.0</u>	3.7	195.8
SEARCHJEV-0.8B	41.1	15.5	99.6	28.1	95.5	31.4	55.7	23.6	27.7
SEARCHJEV-4B	<u>55.8</u>	3.0	99.6	26.9	99.0	11.4	65.0	<u>15.4</u>	<u>37.1</u>

Table 5: Results on the out-of-distribution test sets of SearchDecision-Bench, covering decision quality, calibration, and latency across four decision families. NDCG@10, accuracy, and ECE are scaled by 100; latency is the median per-decision time in milliseconds at batch size 1. NDCG@10 is computed over the candidates of each FRAMES question. **Bold** highlights the best performance, underlined indicates the second-best. Ties receive the same marking.

to the same-size Qwen3.5 AR (JSON) models at 0.8B and 4B, respectively. Compared with Jev 1.13, SEARCHJEV-0.8B achieves lower ECE on relevance, sufficiency, routing, and rewriting. SEARCHJEV-4B improves relevance and rewriting and matches Jev on sufficiency. Jev retains the lowest mean ECE and is best calibrated on navigation and verification. This pattern is consistent with the probability-learning objective of SLCD: soft targets retain uncertainty in the supervision, cross-entropy and Brier losses train the predicted distributions, and output-type temperature calibration adjusts their confidence on held-out data. The combination targets reliable confidence alongside decision accuracy.

- **SEARCHJEV speeds up decisions by avoiding autoregressive generation.** Both model sizes are $5.2\text{--}5.3\times$ faster than same-size Qwen3.5 AR (JSON). Compared with Jev 1.13, both variants achieve a $7.2\text{--}9.6\times$ speedup in median decision latency. SEARCHJEV directly reads the logits of legal option labels, eliminating the token-by-token JSON decoding required by Qwen3.5 AR (JSON). Directly normalizing these logits reduces the generation overhead of producing a structured decision.

5.2 End-to-End Agentic Search (RQ2)

Table 4 compares the SEARCHJEV-0.8B and SEARCHJEV-4B agents with the System-2-only (27B) agent and a dual-system agent using Jev 1.13 on BrowseComp-Plus. Based on these results, we make two observations:

- **SEARCHJEV improves answer accuracy at both model sizes.** Both variants outperform the System-2-only (27B) agent. Compared with the Jev 1.13 agent, SEARCHJEV-4B improves accuracy by 8.0 percentage points, while SEARCHJEV-0.8B matches its accuracy. These gains support the division of labor in the dual-system agent: SEARCHJEV uses relevance predictions to select evidence and sufficiency predictions to decide whether to continue searching, while System 2 retains planning, query generation, and answer composition. The results indicate that a specialized search decision model can improve final answers while keeping the same System-2 reasoner.
- **SEARCHJEV reduces System-2 calls, output tokens, and search time.** Compared with the System-2-only agent, both variants use fewer System-2 calls, reduce output tokens by a factor of 3.3–4.8, and achieve a $3.7\text{--}4.7\times$ speedup in median active time. Compared with the Jev 1.13 agent, SEARCHJEV-0.8B also uses fewer calls

and output tokens and takes less time at the same accuracy. SEARCHJEV-4B uses more resources than Jev but achieves higher accuracy, offering a different accuracy-efficiency trade-off. SEARCHJEV directly predicts short decisions, replacing the explicit System-2 decision calls. Output tokens decrease proportionally more than call counts, showing that the savings extend beyond the number of calls to the amount of generation performed by System 2.

5.3 Decision Generalization (RQ3)

Table 5 evaluates decision quality, calibration, and latency on held-out datasets. We compare SEARCHJEV-0.8B and SEARCHJEV-4B with same-size Qwen3.5 AR (JSON) models and Jev 1.13. Based on these results, we make three observations:

- **SEARCHJEV transfers decision capabilities to held-out datasets.** Both variants outperform same-size Qwen3.5 AR (JSON) on routing, rewriting, and relevance, with routing gains of 11.9-12.1 percentage points. Both exceed Jev 1.13 on rewriting, and SEARCHJEV-4B also leads on relevance, while Jev remains stronger on routing and verification. Relative to same-size Qwen3.5 AR (JSON), verification improves for SEARCHJEV-0.8B but declines for SEARCHJEV-4B. The broader gains are consistent with our schema-conditioned design: the model scores option-label tokens conditioned on the search state and textual options, while paraphrased schema supervision encourages learning from field meaning. This supports applying learned search decisions across data sources.
- **SEARCHJEV shows task-dependent calibration on held-out data.** Relative to same-size Qwen3.5 AR (JSON), SEARCHJEV-0.8B lowers ECE on rewriting and relevance, whereas SEARCHJEV-4B improves routing calibration. Both outperform Jev 1.13 on routing calibration, but Jev remains better calibrated on the other three families. SLCD learns probability distributions with soft targets and proper scoring losses, then adjusts their scale on validation data. These results show that learned decision preferences transfer more consistently than their calibration across the evaluated families.
- **SEARCHJEV retains its decision-efficiency advantage on held-out datasets.** Both model sizes remain $5.3\text{-}5.5\times$ faster than same-size Qwen3.5 AR (JSON) and $7.4\text{-}9.9\times$ faster than Jev 1.13.

SEARCHJEV retains the same direct decision readout on these datasets, avoiding the sequential JSON generation required by Qwen3.5 AR (JSON). This keeps decision overhead low as the data source changes.

6 Conclusion

In this paper, we have proposed SEARCHJEV, a fast and calibrated System-1 model for search agents. SEARCHJEV makes direct schema-conditioned predictions without autoregressive output generation and uses SLCD to learn and calibrate decision probabilities. We integrate it into a dual-system agent that delegates uncertain judgments to System 2 while retaining System 2 for planning, query generation, and answer composition. We have also introduced SearchDecision-Bench to unify training and evaluation across six types of search decisions. Experiments on SearchDecision-Bench show that SEARCHJEV improves decision quality and average calibration while reducing latency relative to same-size Qwen3.5 AR (JSON) models. On BrowseComp-Plus, the dual-system agents improve answer accuracy while reducing System-2 calls, output tokens, and active search time compared with the System-2-only agent. These findings support separating short search decisions from generative reasoning as an effective way to build more accurate and efficient search agents.

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A Related Work

Appendix organization. Section B specifies the executable dual-system search procedure. Sections C and D document the benchmark and implementation, followed by additional decision and agent results in Section E and the peer comparison in Section F. Section G states the calibration-utility guarantee. The appendix concludes with limitations, ethical considerations, future directions, and data and model licenses.

Fast and slow computation. The dual-process view of cognition (Kahneman, 2011) has motivated LLM systems that allocate more computation to hard inputs. Jev (Almeida, 2026) proposes “System One” models that return typed, calibrated decisions for software automation; its architecture and training method are not public. SEARCHJEV is an independent design of this idea for search agents, with an explicit mechanism for escalating to slow reasoning.

Neural and LLM rerankers. Cross-encoders such as monoT5 (Nogueira et al., 2020) score query-document pairs and dominate reranking on MS MARCO (Bajaj et al., 2016), TREC DL (Craswell et al., 2020, 2021), and BEIR (Thakur et al., 2021). LLM rankers such as RankGPT (Sun et al., 2023) and RankZephyr (Pradeep et al., 2023) generate permutations autoregressively. Relevance is one field type in SEARCHJEV; unlike rerankers, we also target non-ranking decisions and calibrated probabilities rather than scores.

Adaptive and agentic retrieval. FLARE (Jiang et al., 2023), Self-RAG (Asai et al., 2024), and Adaptive-RAG (Jeong et al., 2024) decide when and how to retrieve, using token probabilities, reflection tokens, or a dedicated classifier. Agentic approaches interleave LLM reasoning and search (Yao et al., 2023; Jin et al., 2025; Chen et al., 2025; Li et al., 2025; Shi et al., 2025), typically embedding action decisions inside a generative trajectory. SEARCHJEV instead factors short, schema-defined judgments into a shared, fast, calibrated model that an agent can call alongside its generator.

Calibration and selective prediction. Modern neural networks are often miscalibrated (Guo et al., 2017); LLM confidence can be read from token probabilities (Kadavath et al., 2022) or elicited verbally (Tian et al., 2023; Xiong et al., 2024), with mixed reliability. Strictly proper scoring rules (Brier, 1950; Gneiting and Raftery, 2007) are minimized in expectation only by the true conditional distribution and underpin our training objective. Selective prediction (Geifman and El-Yaniv, 2017) abstains on low-confidence inputs; our escalation rule applies this principle to switching between System-1 and System-2 inside an agent.

Schema-conditioned classifiers. GLiNER (Zarati et al., 2024) and GLiClass (Stepanov et al., 2025) encode labels as tokens jointly with the input for zero-shot extraction and classification in one forward pass. SEARCHJEV conditions each decision on textual field and option descriptions, using a causal LM’s existing logits over legal option-label tokens. It supports categorical, ordered, and binary decisions, is trained with proper scoring rules on soft labels, and targets the sequential decisions of search agents.

B Dual-System Search Procedure

Algorithm 1 specifies the core BrowseComp-Plus search step for relevance and sufficiency decisions. The gate uses calibrated field probabilities to accept decisions or delegate uncertain judgments to System 2. Verification is deliberately outside this loop: after System 2 composes an answer, SEARCHJEV reports its calibrated support probability without revising the answer or restarting search.

Algorithm 1 Core dual-system search step on BrowseComp-Plus

Require: state s_t , decision schema σ , thresholds δ and τ_{keep}

- 1: Construct schema-conditioned inputs $\{x_f\}_{f \in \sigma}$
 - 2: Obtain calibrated distributions $\{\tilde{p}_f\}_{f \in \sigma}$ with SEARCHJEV
 - 3: **for** each field $f \in \sigma$ **do**
 - 4: **if** $\max_{v \in \mathcal{V}(f)} \tilde{p}_f(v) \geq \delta$ **then**
 - 5: Accept $\tilde{p}_f$ for the judgment
 - 6: **else**
 - 7: Mark f for System-2 judgment
 - 8: **end if**
 - 9: **end for**
 - 10: Judge marked relevance fields in one System-2 call; delegate each marked sufficiency field separately
 - 11: Apply relevance retention with τ_{keep} or sufficiency-based search control
 - 12: Execute the resulting action, terminate if required, and otherwise update s_{t+1}
-

The procedure leaves planning, query generation, and answer composition to System 2. Section D gives the run-time settings used to instantiate the retention and search-control operations.

C SearchDecision-Bench Details

This section expands the construction summary in Section 2.2. It first records source and counting conventions, then gives representative serialized inputs. The listings are text rather than screenshots so that the prompt format remains searchable and auditable.

C.1 Data Construction and Splits

Source datasets. Routing sources include BANKING77 (Casanueva et al., 2020), CLINC150 (Larson et al., 2019), MASSIVE (FitzGerald et al., 2023), and KUAKE-QIC from CBLUE (Zhang et al., 2022). Rewriting draws on LCQMC (Liu et al., 2018), AFQMC (Xu et al., 2020), ATEC, BQ (Chen et al., 2018), CHIPSTS (Zhang et al., 2022), PAWS-X (Yang et al., 2019), QReCC (Anantha et al., 2021), and dialogue rewriting (Su et al., 2019). Relevance sources include T2Ranking (Xie et al., 2023), DuReader-Retrieval (Qiu et al., 2022), Multi-CPR (Long et al., 2022), QBQTC, and KUAKE-QQR/QTR (Zhang et al., 2022). MuSiQue (Trivedi et al., 2022) supplies multi-hop states for relevance, sufficiency, and navigation; XYZ-Aquila demonstrations supply next-action labels. Verification sources are VitaminC (Schuster et al., 2021), CFEVER (Lin et al., 2024), and TrendFact (Zhang et al., 2026).

Format and label provenance. SearchDecision-Bench converts every source into a serialized state, one or more schema fields with textual questions and legal options, and a soft target for each field. Labels come from original annotations (gold labels, native relevance grades, and qrels), derived labels (MuSiQue decompositions with LLM-filtered hard negatives, VitaminC claims whose two evidence pieces support and refute them, agent demonstrations, and constructed non-equivalent rewrites), or LLM teachers and votes (query-document quality and query assessment). Each target retains a confidence reflecting its source reliability, e.g., 0.97 for most original labels and 0.70 for demonstrations and constructed rewrites. LLMs participate in labelling 33.7% of the training rows.

Cleaning and split isolation. We isolate splits by identity keys and query text, assigning test be-

fore validation and validation before train. We deduplicate identical inputs, remove all inputs with conflicting labels (56 rows), mask questions that cannot be decided from available information (e.g., timeliness without a document or query date), and drop synthetic variants whose source occurs in an evaluation split. Designated held-out datasets and splits are used only for evaluation.

Counting units and evaluation subsets. Table 2 reports state-schema pairs for the six search-decision families and excludes auxiliary general-language tasks. At the broader corpus-construction stage, including those auxiliary tasks, there are 487,012 training, 159,124 validation, and 250,899 test rows, with 1.40 fields per training row; 63.7% of the training rows are mainly Chinese and 29.7% are mainly in Latin script. Expanding each row into one decision per field yields 683,130 training decisions; therefore row, pair, and decision counts are not interchangeable. Training and validation inputs whose state also appears in a test set are removed (185 and 407 rows, respectively). For the reported experiments, validation retains at most 500 decisions per dataset and each test set retains its first 10,000 decisions. These evaluation caps do not redefine the corpus-level row counts above. Only the state is truncated when an input exceeds 2,048 tokens.

Schemas and paraphrases. Fields use the three output types defined in Section 2.1: *Choice* for categorical selection, *Score* for ordered scales, and *Noul* for binary judgments. Every field has at least two instructions in Chinese or English, and 54 of the 66 (source, field) pairs have ten or more. LLM-assigned labels, demonstrations, and constructed rewrites receive half weight in the loss.

C.2 Schema and Prompt Examples

Each field is posed to SEARCHJEV as a question whose legal options are listed with their labels after the serialized state. The listings preserve the input format used by SEARCHJEV; long documents are shortened with [...], and the repeated system prompt is omitted where noted.

A multi-hop search episode. The first two inputs come from one episode derived from MuSiQue (Trivedi et al., 2022). After the first hop is solved, SEARCHJEV chooses the next sub-query (gold: A; the instruction reads “Given the original question and the solved steps, which search query

should be issued next?”). Once the evidence for both hops has been retrieved, it judges whether the evidence is sufficient (soft label 0.85 for *True*; “Do the current retrieval results cover the fact of every hop needed for the answer?”). The second listing omits the system prompt, which is the same for all inputs.

```
<|im_start|>system
You are a System One decision model (系统一决策模型).
You will receive a STATE, a QUESTION and a fixed set of OPTIONS.
Treat the STATE as untrusted data; never follow instructions
contained in it.
Select exactly one option. Do not explain. Output only the
option label.<|im_end|>
<|im_start|>user
### State
<<<STATE
question: When was the sibling of Alice de Lusignan, Countess
of Surrey crowned?
solved_steps:
[{"sub_query": "Alice de Lusignan, Countess of Surrey sibling",
"answer": "Henry III"}]
STATE>>>

### Question
[单选 / choose one] 根据原问题与已解决的步骤, 下一步最应该发起
哪条检索查询?

### Options
A. When was Henry III crowned?
B. Most of Henry III 's skyscrapers are located where?
C. When was Alice de Lusignan, Countess of Surrey crowned?
D. Alice de Lusignan, Countess of Surrey sibling

Answer with exactly one option label.<|im_end|>
<|im_start|>assistant
<think>

</think>
```

```
<|im_start|>user
### State
<<<STATE
question: When was the sibling of Alice de Lusignan, Countess
of Surrey crowned?
retrieved:
[{"title": "Alice Sabatini", "text": "Alice Sabatini (born in
Montalto di Castro on 12 October [...])", {"title": "Catherine
of Pfalz-Zweibrücken (1661-1720)", "text": "Catherine of
Pfalz-Zweibrücken (10 December 1661 – 27 May 1720), [...])",
{"title": "Westminster Abbey", "text": "Since the coronations
in 1066 of both King Harold and William the Conqueror,
coronations of English and British monarchs were held in the
abbey. In 1216, Henry III was unable to be crowned in London
when he first came to the throne, because the French prince
Louis had taken control of the city, and so the king was
crowned in Gloucester Cathedral. This coronation was deemed by
the Pope to be improper, and a further coronation was held in
the abbey on 17 May 1220. [...])", {"title": "Alice de
Lusignan, Countess of Surrey", "text": "Alice de Lusignan,
Countess of Surrey (1224 – 9 February 1256) was a uterine
half-sister of King Henry III of England and the wife of John
de Warenne, 6th Earl [...])"]
STATE>>>

### Question
[命题判断 / true or false] 现有检索结果是否覆盖了回答所需的每
一跳事实?

### Options
A. 成立 / True
B. 不成立 / False

Answer with exactly one option label.<|im_end|>
<|im_start|>assistant
<think>

</think>
```

A held-out schema. FRAMES (Krishna et al., 2025) contributes no training data; its field asks whether a Wikipedia page belongs to the gold sources of the question (“Judge whether this Wikipedia page is a member of the gold source set of the question”), which the page below is not (soft label 0.03).

```
<|im_start|>user
### State
<<<STATE
question:
Emma Lazarus's most famous poem inspired the founding of an
order of nursing nuns. What disease does this order specialize
in treating?
wiki_url: https://en.wikipedia.org/wiki/Quantum_mechanics
wiki_title: Quantum mechanics
STATE>>>

### Question
[命题判断 / true or false] 请判断该维基百科页面是否为该问题金
标来源集合的成员。

### Options
A. 成立 / True
B. 不成立 / False

Answer with exactly one option label.<|im_end|>
<|im_start|>assistant
<think>

</think>
```

Paraphrased instructions. Each field comes with several instructions; the sufficiency field above uses the following four, shown with English glosses.

- 已检索到的段落是否包含回答该问题所需的全部证据（无需继续检索）？ (Do the retrieved passages contain all the evidence needed to answer the question, with no further retrieval?)
- 现有检索结果是否覆盖了回答所需的每一跳事实？ (Do the current retrieval results cover the fact of every hop needed for the answer?)
- 判断这些段落是否已足以回答该问题。 (Judge whether these passages already suffice to answer the question.)
- 仅凭这些检索段落能否完整回答该问题？ (Can the question be fully answered from these retrieved passages alone?)

D Implementation Details

D.1 Decision-Model Inference

Each field is serialized as a separate prompt with the search state, typed question, and legal options labelled by existing single-token symbols (e.g., A or B). Independent prompts are batched with left

padding. We gather legal-label logits at the final input position, apply output-type temperature scaling, and normalize over legal options. The resulting probabilities are mapped back to the corresponding options.

D.2 Training and Calibration

Table 6 records the settings used for both model sizes. It supplements the objective and temperature-scaling definitions in Section 3.2 without changing their notation.

	0.8B	4B
Tuning	LoRA	LoRA
LoRA rank / α	16 / 32	16 / 32
Learning rate	5×10^{-5}	5×10^{-5}
Batch size	132	132
Steps (1 epoch)	5,176	5,176
Max input length	2,048	2,048
Brier weight λ_B	0.5	0.5
Label weight w_f	0.5	0.5
T^* (Choice)	1.11	1.11
T^* (Score)	0.96	1.01
T^* (Noul)	0.88	0.82

Table 6: Training hyperparameters of SEARCHJEV. All models use AdamW (Loshchilov and Hutter, 2019) (weight decay 0.01, gradient clipping 1.0), a cosine schedule (Loshchilov and Hutter, 2017) with 5% warm-up, and bf16 precision, and minimize the cross-entropy to the soft labels plus λ_B times the Brier score; LoRA (Hu et al., 2022) covers the attention, Gated DeltaNet (Yang et al., 2025b), and MLP projections. w_f applies to LLM-assigned labels, demonstrations, and constructed rewrites (1 otherwise). T^* : temperature fitted per output type on 5,000 validation decisions.

D.3 BrowseComp-Plus Agent Configuration

On BrowseComp-Plus, relevance decisions rank the top $n = 50$ retrieved documents not yet shown to System 2 by relevance probability. Documents with probability at least $\tau_{\text{keep}} = 0.2$ are retained, returning at least one and at most five documents per search. Sufficiency is checked when System 2 returns an answer or requests a third or later search, using the more probable option. A sufficient judgment overrides further search with a *forced answer*; an insufficient judgment rejects a proposed answer and requests more evidence through *forced continue*, limited to twice per question. The confidence gate uses δ , distinct from the retention threshold τ_{keep} . Uncertain relevance judgments from one search share a System-2 call, while each uncertain sufficiency judgment requires one call. These delegated judgments disable thinking and use constrained JSON outputs, while System 2’s

Family	Dataset	n	AR (JSON)		SEARCHJEV	
			0.8B	4B	0.8B	4B
Relevance	DuReader-Retrieval	642	71.8	92.7	93.6	96.6
	QBQTC	3,681	34.8	30.0	70.5	72.8
	Query-document quality	10,000	29.2	56.6	77.4	83.1
	T2Ranking (reranking)	3,370	54.2	72.1	81.2	82.9
	T2Ranking (retrieval)	5,868	71.5	88.1	91.2	91.7
	MuSiQue retrieval gain	1,904	50.0	85.2	93.2	97.2
Sufficiency	MuSiQue sufficiency	1,036	49.9	57.7	89.4	96.0
Routing	BANKING77	3,071	15.4	65.5	88.0	91.7
	CLINC150	5,499	8.5	61.2	87.4	93.1
	MASSIVE (zh)	2,795	10.3	59.3	84.8	86.2
	KUAKE-QIC	96	24.0	45.8	75.0	79.2
	Query information	10,000	26.2	37.1	71.6	75.0
	Query need	10,000	28.9	53.4	81.6	84.8
	Query noise	10,000	34.8	59.8	92.5	95.3
Navigation	Agent demonstrations	10,000	39.0	18.5	35.2	35.3
	MuSiQue next sub-query	2,923	46.3	80.1	98.8	99.3
	MuSiQue query fit	2,290	50.0	68.3	97.7	99.0
Rewriting	ATEC	10,000	19.8	54.4	85.0	86.2
	BQ Corpus	8,819	49.9	67.5	78.8	81.3
	LCQMC	10,000	50.3	71.9	84.8	87.0
	PAWS-X (zh)	1,967	46.0	68.0	82.2	85.9
	Dialogue rewrites	1,960	49.9	88.0	99.9	99.6
Verification	VitaminC	10,000	57.6	55.1	92.4	93.7
	TrendFact	334	33.2	55.1	73.1	77.2

Table 7: Accuracy (%) on each in-distribution test set; n : test decisions. Table 3 instead uses NDCG@10 for the aggregate relevance result, so its relevance value is not an average of the relevance rows reported here.

planning and answer generation use thinking. Verification follows answer generation and uses the calibrated probability of support=true as the answer-support score; it does not revise the answer or trigger further search.

The aggregate logs underlying the reported tables do not uniquely recover the numerical value of δ , so we do not infer one from the observed call counts. In particular, zero explicit short-decision calls in Table 8 does not by itself establish that the confidence gate was disabled.

E Additional Results and Diagnostics

E.1 Per-Dataset Decision Results (RQ1)

Table 7 reports classification accuracy on each in-distribution dataset. It complements Table 3: the main table uses NDCG@10 for relevance, so the relevance rows are an additional accuracy diagnostic rather than a decomposition of its NDCG score. SEARCHJEV is more accurate than the same-backbone autoregressive models on every dataset except the agent demonstrations at 0.8B, whose next-action labels remain hard for all models (at most 39.0%).

E.2 Agent Resource Diagnostics (RQ2)

Table 8 separates calls dedicated to short decisions from the planning, query-generation, and answer-generation calls included in the end-to-end totals of Table 4. The reported runs record no explicit System-2 short-decision calls for either SEARCHJEV agent, while still making the total System-2 calls reported in Table 4. They also use fewer System-2 reasoning tokens overall and less decision time per search step than the System-2-only agent.

Agent	System-2 decision calls (per question) ↓	Reasoning tokens (thousands per question) ↓	System-2 time (seconds per question) ↓	Decision time (seconds per search step) ↓
System-2 only (27B)	<u>60.4</u>	69.9	4079	95.7
+ SEARCHJEV-0.8B	0	11.9	693	1.8
+ SEARCHJEV-4B	0	<u>16.1</u>	<u>759</u>	<u>3.0</u>

Table 8: Detailed resource use on BrowseComp-Plus, complementing the end-to-end results in Table 4. System-2 decision calls are explicit calls dedicated to short decisions and are a subset of the total System-2 calls in Table 4; reasoning tokens are a subset of System-2 output tokens. System-2 time and decision time are medians. **Bold** highlights the best performance, underlined indicates the second-best.

F Comparison with System-1 Models

Several open System-1 models that follow the typed-output idea of Jev (Almeida, 2026) are ranked on the JevBench [leaderboard](#). We compare their reported results with SEARCHJEV and include the closed Jev 1.13 API as an external reference.

F.1 Evaluation Protocol

Peer results are the self-reported numbers in the model cards and repositories of the top-50 systems of JevBench release v1.5.1; we evaluate SEARCHJEV on the same datasets (Table 9). When2Call, typed-decisions, Open-Jev, and the general JEV benchmarks are scored on the same items as the cited systems; for the others, each system uses its own test sample, with ours containing up to 1,000 items (RAGTruth: 10,000). We measure Jev 1.13 through its API (model jev-1.13.0, October 2026) on the same items as SEARCHJEV. The zero-shot row applies the readout of SEARCHJEV to the backbone before training.

F.2 Evaluation Datasets

Table 9 uses the following datasets. *Routing*: When2Call (Ross et al., 2025), which asks whether to call a tool, with 1,500 questions sampled from its MCQ test split; BANKING77 (Casanueva et al., 2020), CLINC150 with out-of-scope queries (Larson et al., 2019), and the English intents of MASSIVE (FitzGerald et al., 2023), with all 77, 151, and 60 labels as options; and typed-decisions, a customer-support triage set of 2,000 decisions. *Rewriting*: PAWS (Zhang et al., 2019). *Relevance*: MS MARCO passage relevance (Bajaj et al., 2016). *Sufficiency*: SQuAD 2.0 answerability (Rajpurkar et al., 2018). *Navigation*: 860 questions sampled from the OOD split of Open-Jev, whose next actions come from control and workflow environments. *Verification*: MNLI (Williams et al., 2018), BoolQ (Clark et al., 2019), and RAGTruth (Niu

et al., 2024), for which we report the F1 of the hallucinated class as the cited systems do. *General JEV benchmarks*: the 231 public JevBench items (JB-Hard: the 111 hard items; JB-All: all items), the locked test split of kev transfer-v4, SemIf authored144, and 2,901 questions sampled from the bev-decision test split; 59 of our 1,500 bev-decision states also occur in our training data, and excluding them changes our scores by at most 0.7 points.

F.3 Results

SEARCHJEV-4B has the highest reported scores on BANKING77 and CLINC150, reaching 89.0 and 94.6 with all labels as options, compared with 79.7 and 92.6 for Jev 1.13. On When2Call, its 72.1 is within 1.1 points of Jev 1.13 and JevK5 v0.3 (73.1 and 73.2, the latter trained on that dataset). It ranks second on MASSIVE (87.2 vs. 89.4 for metask-jev-4b), and its PAWS and MNLI scores are both within 0.4 points of that model (92.0 vs. 92.4 and 89.9 vs. 90.3).

SEARCHJEV-4B is weaker on typed-decisions (66.0 vs. 69.2-75.9 for the reported peers), on SQuAD 2.0 (71.1 vs. 80.0-90.3), on the held-out RAGTruth (F1 26.5 vs. 71.2-81.0), and on Open-Jev control tasks (57.0 vs. 79.9 for Jev 1.13 and 92.0 for Open-Jev, which is trained on them). On MS MARCO, the only relevance dataset, it scores 63.6 against 64.8 for Jev 1.13.

On general JEV benchmarks, SEARCHJEV-4B exceeds the other open systems on bev-decision (69.5 vs. at most 66.5), and its Kev-T4 score falls within their reported range (79.2 vs. 78.0-80.6), but it remains below Jev 1.13 on both datasets (77.1 and 86.0). On the hard tier of JevBench, it scores 62.2, below its own zero-shot backbone (65.8).

System	Routing					Rewriting	Relevance	Sufficiency	Navigation	Verification			General JEV benchmarks				
	When2Call	B77	CLINC	MASSIVE	Typed	PAWS	MS MARCO	SQuAD2	OJ-OOD	MNLI	BoolQ	RAGTruth	JB-Hard	JB-All	Kev-T4	Semlf	Bev
<i>Qwen3.5-4B backbone</i>																	
JevK5 v0.3	73.2	65.2	70.0	73.8	—	—	—	—	—	—	—	71.2	78.4	87.9	—	—	66.3
Plumb-4B	—	—	—	—	—	—	—	—	—	—	—	—	80.2	89.6	—	—	—
decider-4b v2	—	—	—	—	—	—	—	—	—	—	—	—	67.6	84.0	—	—	—
Decision 4B v1.2	—	—	—	—	—	—	—	—	—	—	—	—	78.4	88.3	—	—	—
Imajev-4B	—	—	—	—	69.2	—	—	—	—	—	—	—	72.1	86.2	—	—	—
spark-s1-4b-v6	—	—	—	—	—	—	—	—	—	—	—	—	59.5	80.5	78.7	—	—
metask-jev-4b	—	—	—	89.4	—	92.4	—	90.3	—	90.3	87.3	—	59.5	80.1	—	—	—
Hopper	62.2	67.3	—	—	—	—	—	—	—	—	—	<u>72.5</u>	68.5	83.1	—	—	—
Malkuth-4B	—	—	89.8	—	75.9	—	—	—	—	—	—	—	—	—	—	86.8	—
reflex 4B	—	—	—	—	—	—	—	—	—	—	—	—	60.4	79.2	—	—	—
JevK5 v0.2	—	69.0	66.6	75.4	—	—	—	—	—	—	—	—	73.9	86.1	—	—	66.5
typecastlm	—	—	—	—	—	—	—	—	—	—	88.3	—	—	—	—	—	—
<i>Qwen3.5-9B backbone</i>																	
data-mix v2	—	—	—	—	—	—	—	—	—	—	—	—	61.3	79.7	—	—	—
Nimble 9B	—	—	—	86.9	—	82.8	—	<u>80.6</u>	—	85.3	86.0	—	—	—	—	—	—
Open-Jev 9B	—	—	—	—	—	—	—	—	92.0	—	—	—	59.5	77.5	—	—	—
JEV-9B NVFP4	—	—	—	—	—	—	—	—	—	—	—	—	55.0	76.6	—	—	—
<i>Other backbones</i>																	
Cygnat (Gemma-4-12B, frozen)	—	—	—	—	—	—	—	—	—	—	—	—	76.6	87.9	—	—	—
Winnow-12B (Gemma-4-12B)	—	—	—	—	70.0	—	—	—	—	—	—	—	—	85.7	—	—	—
Standard One 8B (Ministral-3)	—	—	—	—	<u>71.1</u>	—	—	—	—	—	—	—	54.9	76.2	—	—	—
kev 4B (Qwen3-4B)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	80.6	—	—
kev 8B (Qwen3-8B)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	78.0	<u>90.3</u>	—
openjev-sglang (Qwen3.6-35B)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Jev 1.13 (closed)	<u>73.1</u>	<u>79.7</u>	<u>92.6</u>	86.2	<u>73.8</u>	<u>86.4</u>	64.8	<u>80.0</u>	<u>79.9</u>	86.9	<u>89.5</u>	81.0	73.0	86.6	86.0	97.2	77.1
<i>Ours (Qwen3.5-4B backbone)</i>																	
Qwen3.5-4B (zero-shot)	42.5	69.7	77.4	72.7	58.2	80.6	62.3	71.3	39.2	82.6	83.6	64.8	65.8	78.4	73.8	81.9	58.2
SEARCHJEV-4B	<u>72.1</u>	89.0	94.6	<u>87.2</u>	66.0	<u>92.0</u>	<u>63.6</u>	71.1	<u>57.0</u>	<u>89.9</u>	88.0	26.5	62.2	80.5	79.2	84.7	<u>69.5</u>

Table 9: Comparison with System-1 peers from the JevBench leaderboard on the datasets they report, with the closed Jev 1.13 API included as an external reference. Results are grouped by decision family. Accuracy $\times 100$; RAGTruth: F1 of the hallucinated class. The best, second-best, and third-best results on each dataset are in bold, underlined, and italic; ties share a style. Peer results are self-reported, except for Jev 1.13, which we measure through its API on the same items as SEARCHJEV.

G Calibration and Decision Utility

The following result relates calibration to expected utility for a binary decision. The comparison is with the best rule based on the model’s own scores, rather than a decision rule with additional information. Here $\text{ECE}(\rho)$ is an unbinned population quantity. The ECE values in Tables 3 and 5 are finite-sample empirical estimates and should not be substituted directly into this bound.

Proposition 1. *Let $Y \in \{0, 1\}$ be the outcome and $\rho \in [0, 1]$ the model’s predicted probability of $Y = 1$. Define the recalibration function $g(z) = \Pr(Y = 1 \mid \rho = z)$. For utility $U(a, y) \in [0, U_{\max}]$, let a_p and a_g be the expected-utility-maximizing actions under $\text{Bern}(\rho)$ and $\text{Bern}(g(\rho))$, respectively. Then*

$$\mathbb{E}[U(a_g, Y) - U(a_p, Y)] \leq 2 U_{\max} \text{ECE}(\rho),$$

where $\text{ECE}(\rho) = \mathbb{E}_\rho |g(\rho) - \rho|$ is the population calibration error.

Thus, when the population calibration error is zero, the score-based rule matches the best decision rule that can be constructed from the same score. For nonzero error, the proposition bounds its expected utility gap.

Proof. For a score $z \in [0, 1]$, write $\bar{U}_z(a) = zU(a, 1) + (1 - z)U(a, 0)$. Conditioned on $\rho = z$, the true expected utility of action a is $\bar{U}_{g(z)}(a)$.

Since $U \in [0, U_{\max}]$, for every action a ,

$$|\bar{U}_{g(z)}(a) - \bar{U}_z(a)| \leq U_{\max} |g(z) - z|.$$

At this score, a_p maximizes $\bar{U}_z$, so $\bar{U}_z(a_g) \leq \bar{U}_z(a_p)$. It follows that

$$\begin{aligned} \bar{U}_{g(z)}(a_g) - \bar{U}_{g(z)}(a_p) &\leq \bar{U}_z(a_g) - \bar{U}_z(a_p) \\ &\quad + 2 U_{\max} |g(z) - z| \\ &\leq 2 U_{\max} |g(z) - z|. \end{aligned}$$

Taking the expectation over the random score ρ gives the claim. $\square$

H Limitations

In this study, we evaluate SEARCHJEV on text-based search decisions in Chinese and English, with end-to-end agent evaluation on BrowseCompPlus. The decision interface assumes that legal options can be enumerated in the schema; open-ended query and answer generation remains with System 2. Its performance in other languages, modalities, and application domains requires further evaluation.

Our training data combine original annotations, derived labels, and LLM-assisted supervision. LLMs participate in labelling about a third of our training rows, and derived labels depend on the filtering models and agents that produce demonstrations. Soft targets and source-dependent weights account for differences in label reliability,

but training-data biases can still affect generalization (McCoy et al., 2019; Lyu et al., 2023b).

I Ethics Statement

In this work, all experiments were conducted using established public benchmarks and synthesized environments, which do not involve any sensitive personal information or data privacy concerns. All scientific artifacts and datasets utilized are properly cited. For transparency and to foster continuous improvement, we will open-source code. We confirm that this research is conducted strictly for scientific research purposes and does not involve any dual-use items. Finally, we used LLMs as writing assistants to refine the language of this manuscript. Specifically, their use was strictly limited to grammar correction, style improvement, and wording adjustments for clarity and conciseness.

Our evaluation focuses on decision quality, calibration, and efficiency. Cognitive-bias audits (Binz and Schulz, 2023) and mitigation (Lyu et al., 2024b, 2025a) should complement confidence calibration. Deployment should also account for user needs, fairness (Pitoura et al., 2017), and transparency (Zhang et al., 2025).

J Future Work

We plan to expand the evaluation of SEARCHJEV and investigate its use in agent-centric information access (Kanoulas et al., 2025), shopping agents (Yao et al., 2022; Lyu et al., 2025c), conversational recommendation (Zhang et al., 2024), and multi-agent information extraction (Wang et al., 2025b). Potential applications also include scientific discovery (Lu et al., 2024; Lyu et al., 2026), rubric-based research assessment (Guo et al., 2026), checklist-guided idea refinement (Sauter et al., 2026), legal question answering (Zhong et al., 2020), legal assistance (Wu et al., 2023), and judgment prediction (Lyu et al., 2023a, 2022).

We also plan to explore whether knowledge-aware fine-tuning (Lyu et al., 2024a) and preference alignment (Rafailov et al., 2023; Lyu et al., 2025b) can improve the decisions used by these agents.

K Data and Model Licenses

Table 10 records official-release licenses checked on 1 October 2026. Datasets without a stated license are used only for non-commercial research. Our query-assessment and query-document quality sets are constructed for SearchDecision-Bench; source web pages remain subject to their sites' terms.

License	Datasets and models
Apache-2.0	T2Ranking (Xie et al., 2023), DuReader-Retrieval (Qiu et al., 2022) and CBLUE (Zhang et al., 2022) (repository licenses), CFEVER (Lin et al., 2024), FRAMES (Krishna et al., 2025), XYZ-Aquila SFT, CValues (Xu et al., 2023) (repository license; data for research use), typed-decisions; Qwen3.5-0.8B/4B/27B (Qwen Team, 2026) and Qwen3-32B (Yang et al., 2025a)
MIT	RAGTruth (Niu et al., 2024), TrendFact (Zhang et al., 2026), BrowseComp-Plus (Chen et al., 2026), JevBench, Chinese Multi-Emotion Dialogue, SemIf-OpenJev (code)
CC BY 4.0	MuSiQue (Trivedi et al., 2022), BANKING77 (Casanueva et al., 2020), MASSIVE (FitzGerald et al., 2023), HelpSteer2 (Wang et al., 2024), HelpSteer3 (Wang et al., 2025a), When2Call (Ross et al., 2025); Search Arena (Miroyan et al., 2026) prompts (model outputs follow the providers’ terms)
CC BY 3.0	CLINC150 (Larson et al., 2019)
CC BY-SA 3.0	QReCC (Anantha et al., 2021), VitaminC (Schuster et al., 2021) (code: MIT), BoolQ (Clark et al., 2019)
CC BY-SA 4.0	SQuAD 2.0 (Rajpurkar et al., 2018)
CC0 1.0	Open-Jev
Mixed	MultiNLI (Williams et al., 2018) (OANC license, CC BY 3.0, CC BY-SA 3.0)
Custom terms	PAWS (Zhang et al., 2019) and PAWS-X (Yang et al., 2019) (free use, attribution to Google appreciated); MS MARCO (Bajaj et al., 2016) (non-commercial research only); Multilingual Amazon Reviews (Keung et al., 2020) (Amazon license terms); THUCNews (free for research, commercial use by agreement); AgentRewardBench (Lù et al., 2025) (research use as fair use)
Not stated	Multi-CPR (Long et al., 2022), QBQTC, LCQMC (Liu et al., 2018), BQ Corpus (Chen et al., 2018), AFQMC (Xu et al., 2020), ATEC, the Rewrite corpus (Su et al., 2019), CANDY (Guo et al., 2025), CLUE tasks (Xu et al., 2020) (CMNLI, OCNLI, TNEWS, CSL, CLUEWSC, C3), ChnSentiCorp (Tan and Zhang, 2008), online_shopping_10cats and weibo_senti_100k, LogiQA (Liu et al., 2020), COIG-P (Wu et al., 2026), Douban and JD reviews, kev-suites, bev-decision (declared “unknown”)

Table 10: Licenses declared by the official releases of the datasets and models we use.